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Corresponding Author	FamilyName	**Zhang**
	Particle	
	Given Name	**Lihua**
	Suffix	
	Division	Academy for Engineering and Technology
	Organization	Fudan University
	Address	Shanghai, 200433, China
	Phone	
	Fax	
	Email	lihuazhang@fudan.edu.cn
	URL	
	ORCID	
Author	FamilyName	**Feng**
	Particle	
	Given Name	**Jinghui**
	Suffix	
	Division	Academy for Engineering and Technology
	Organization	Fudan University
	Address	Shanghai, 200433, China
	Division	College of Mechatronics
	Organization	Changchun Polytechnic
	Address	Changchun, 130022, China
	Phone	
	Fax	
	Email	
	URL	
	ORCID	
Author	FamilyName	**Zhang**
	Particle	
	Given Name	**Xukun**
	Suffix	
	Division	Academy for Engineering and Technology
	Organization	Fudan University
	Address	Shanghai, 200433, China
	Phone	
	Fax	
	Email	
	URL	
	ORCID	
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Abstract

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Keywords (separated by '-')

Bald eagle search - Meta-heuristic algorithm - Feature selection - Classification - Optimization

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RESEARCH ARTICLE

Improved Bald Eagle Search Optimization Algorithm for Feature Selection in Classification

Jinghui Feng[1,2] · Xukun Zhang[1] · Lihua Zhang[1]

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Abstract

Feature selection serves as an effective way for decreasing the quantity of features within a dataset, which helps enhance the performance of classification in machine learning (ML). In this paper, we formulate a joint feature selection problem to reduce the number of selected features while improving the classification accuracy. We propose an improved bald eagle search (IBES) algorithm to solve the optimization problem. Specifically, the BES algorithm is enhanced by introducing the lévy flight mechanism in the selection phase to improve the global search capability of the algorithm. In addition, we adopt an adaptive weighting factor to balance the global and local search capabilities. Finally, a novel mutation mechanism incorporating Gaussian and differential mutation is proposed, which contributes to maintain the diversity of the population. Comparative experiments are conducted with six benchmark algorithms on eighteen typical datasets. The results analysis indicates that the proposed IBES algorithm can achieve higher classification accuracy with a small number of features.

Keywords Bald eagle search · Meta-heuristic algorithm · Feature selection · Classification · Optimization

1 Introduction

With its robust predictive prowess and adept pattern recognition capabilities, machine learning (ML) is extensively applied across diverse domains, such as disease diagnosis in healthcare [1], risk management in finance [2], and user behavior prediction in marketing [3]. In these fields, ML has demonstrated excellent performance [4]. As the big data era unfolds, the advent of extensive datasets has reshaped the landscape of ML [5]. Although these datasets have great potential for uncovering hidden patterns and insights, traditional ML algorithms frequently face a number of challenges and limitations in such context. First, ML algorithms typically face tremendous computational and storage pressure. Second, large-scale datasets lead to varied data quality and the challenge of high dimensionality, which affects the analytical ability of ML algorithms. Therefore, how to efficiently handle large-scale datasets is an important issue facing the current ML field.

To address the challenges posed by large-scale datasets, researchers have proposed various approaches, including distributed computing frameworks [6, 7], parallel processing techniques [8, 9], and algorithm optimization. Among them, feature selection [10] methods stand out due to high efficiency and ease of deployment. Specifically, feature selection involves identifying the utmost relevant attributes from the initial feature set, aiming to diminish data dimensionality and decrease model complexity [11]. In addition, it promotes better interpretability of the models to facilitate a deeper understanding of underlying data patterns. Therefore, feature selection assumes paramount significance, especially within the realm of large-scale datasets, primarily to alleviate computational burdens and mitigate the potential of overfitting.

Filter-based methods and Wrapper-based methods are the two primary types for feature selection [12]. Specifically, Filter-based methods operate independently of subsequent learning algorithms and assess the importance of features through relationships between features or statistical methods [13]. Wrapper-based methods employ classifiers to assess the features obtained through the feature selection process, incorporating feedback from the classifier to guide the selection process. [14]. Clearly, filter-based feature selec-

✉ Lihua Zhang
lihuazhang@fudan.edu.cn

[1] Academy for Engineering and Technology, Fudan University, Shanghai 200433, China

[2] College of Mechatronics, Changchun Polytechnic, Changchun 130022, China

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tion methods have lower computational costs and faster speeds. Nevertheless, wrapper-based methods are more popular because they directly interact with the chosen classifier, which helps improve the accuracy of classifier decisions [15].

It is worth noting that exhaustive and exact search methods become impractical for feature selection on high-dimensional datasets [16], as they encounter ultra-high computational complexity and time consumption. Fortunately, meta-heuristic swarm intelligence algorithms [17] can solve the wrapper-based feature selection problem well. Swarm intelligence algorithms [18] typically find the optimal solution by simulating the exchange of information and cooperative behavior among natural groups and individuals. Typical swarm intelligence algorithms include genetic algorithms (GA) [19, 20], firefly algorithms (FA) [21, 22] and particle swarm optimization (PSO) [23, 24]. These algorithms can effectively explore the feature space and choose a better feature subset in a relatively short time through iteration and evolution.

Among various swarm intelligence algorithms, the Bald Eagle Search (BES) algorithm [25] has emerged as a recent method widely utilized for addressing diverse optimization problems, exhibiting notable efficacy. However, it still faces certain limitations, including challenges in maintaining population diversity and susceptibility to convergence toward local optima. Furthermore, the no-free-lunch (NFL) theorem [26] states that there exists no single feature selection algorithm capable of consistently identifying the best subset of features, and no optimization algorithm can universally outperforms all others. Therefore, the conditions above motivate us to enhance the original BES algorithm, aiming to enhance its capability to address the feature selection problem for extracting the features of higher quality

The contributions of our paper can be summarized as follows:

- *Problem Formulation*: To strike a balance between reducing the dimensionality of the feature space and preserving the discriminative power of the selected features, we formulate a multi-objective joint optimization problem for feature selection, *i.e.*, minimizing the number of selected features while maximizing the classification accuracy.
- *IBES-based Wrapper Feature Selection Algorithm*: we propose a wrapper feature selection algorithm based on improved BES (IBES) algorithm. Specifically, we first introduce Lévy flight mechanism to enhance the explorability of original BES algorithm. Secondly, we design an adaptive weighting operator to balance the global and local search capabilities of the algorithm. Finally, a novel mutation mechanism, which consists of Gaussian mutation and differential mutation, is proposed to improve population diversity.
- *Experiments and Analysis*: Our proposed IBES algorithm has been verified to outperform the original BES as well as other benchmark algorithms in feature selection across 16 typical datasets. Moreover, the numerical results demonstrate the commendable performance of our proposed IBES algorithm, particularly in handling high-dimensional datasets.

The rest of this paper is organized as follows: the related work is introduced in Sect. 2. The Problem Formulation of feature selection is shown in Sect. 3. The improved IBES algorithm is proposed in Sect. 4. The experiments results and analysis are presented in Sect. 5. The conclusion of this paper is summarized in Sect. 6.

2 Related Work

Some existing works utilized the traditional exhaustive search method [27, 28] to address feature selection challenges. However, the computational demands of this method become prohibitive when handling high-dimensional datasets, rendering it impractical. Therefore, some researchers investigated the application of heuristic search methods for feature selection. For example, considering that backtracking algorithm is not scalable to large datasets, the authors in [29] proposed a heuristic-based algorithm in rough sets. Moreover, a two-layer cutting plane algorithm was designed for optimizing multivariate measures in multiple-instance learning problems [30]. In addition, two heuristic-based algorithms, Xvariance and Mutual Congestion, were introduced [31]. In this case, experimental results had demonstrated that Xvariance can effectively address standard datasets, while Mutual Congestion can notably enhance classification accuracy for datasets with high dimensionality.

Moreover, the researchers also investigated to adopt meta-heuristic algorithms to solve the problems of feature selections, especially swarm intelligence algorithms. For example, a method based on GA was proposed for both parameter optimization and feature selection. [32, 33]. In addition, the authors in [34] introduced an approach for feature selection combining rough sets and PSO, which can augment the search capabilities, particularly in medium- or high-dimensional datasets. A novel hybrid classifier model integrating SVM and Ant Colony Optimization (ACO) was introduced. [35], which utilized small and appropriate feature subsets to improve classification accuracy. In addition, The authors introduced an approach for enhancing disease diagnosis accuracy on medical datasets through feature selection utilizing Artificial Fish Swarm Optimization (AFSO). [36]. Moreover, a binary CS method was proposed to detect theft in power distribution systems [37]. In addition, the authors in [38] proposed a whale optimization algorithm

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(WOA)-based wrapper algorithm and used several evaluation metrics to judge its effectiveness, with results indicating its superiority over multiple optimizers. However, these algorithms mentioned above are based on primitive metaheuristic search algorithms which still have the potential to be improved in terms of performance.

Therefore, enhanced swarm intelligent-based methods have received much attention in recent years. To enhance feature selection, the authors in [39] introduced an advanced Crow Search Algorithm (CSA), integrating adaptive awareness probabilities, dynamic exploration of local neighborhoods, and a new global search strategy. In addition, an improved adaptive PSO algorithm was proposed [40], where a heuristic mechanism with directional information was used to enhance the convergence speed, as well as an improved state transition probability rule significantly improved the search efficiency. Moreover, an enhanced PSO algorithm, incorporating lévy flights and a unique mutation mechanism, was introduced, improving its search capability. [41]. Additionally, an improved dragonfly algorithm (DA) which adopted a new evolutionary population dynamics strategy and crossover were introduced to bolster the exploitation ability [42]. The authors in [43] introduced a dynamic opposite learning mechanism into the moth flame optimization algorithm to address the problems of premature convergence and local optima. Moreover, a wrapper-based feature selection algorithm, leveraging a hybrid chaotic binary DA integrated with KNN classifier, was proposed [44]. The authors combined the CS algorithm with a local search technique to obtain the satisfactory feature subset [45]. Additionally, an innovative multi-label correlated redundant feature selection approach grounded on ACO was initially introduced [46]. Here, the initial pheromone for each ant was determined by the normalized cosine similarity between features and class labels, aiming to speed up the convergence. A binary gray wolf optimization (GWO) algorithm was introduced [47], where two novel methods of mapping continuous solution space to discrete solution space were designed to bolster GWO in feature selection. In addition, an enhanced SSA algorithm was presented [48], incorporating inverse learning and lévy flight mechanism to argument the process of initializing population and enhance the global search ability in subsequent iterations.

Note that the studies above improve the performance of the algorithm by enhancing the original swarm intelligence algorithm. Moreover, another approach to enhance performance is to integrate multiple swarm intelligence algorithms to achieve synergistic effects greater than individual parts. For example, the authors introduced a hybrid feature selection algorithm that merges GA and ACO algorithms [49], and this hybrid algorithm fully utilized the advantages of ACO and GA to obtain faster and better search capabilities. Additionally, the integration of simulated annealing (SA) with GWO was employed to pursue optimal solutions at the conclusion of every iteration, with the objective of discerning superior solutions. [50]. Furthermore, a novel modified hybrid version of GWO and PSO with a Tent chaotic mapping-based population initialization process was proposed to mitigate the susceptibility to local optima [51], which can sustain population diversity and bolster global search ability. Moreover, a hybrid technique combining GWO and PSO was proposed, which identified the key features and eliminated the complexity at the same time [52].

Furthermore, integrating other learning mechanisms into swarm intelligence algorithms is also an important way to enhance performance. For example, the authors in [53] used emotional learning mechanism to enhance the population diversity of the social spider optimization algorithm, which significantly improved the search performance. Moreover, an improved chimp optimization algorithm was proposed [54], where a tracking learning behavior was designed to enhance search accuracy, and a fuzzy opposite learning was employed to strengthen global search capability. In addition, the authors in [55] proposed a highly innovative hybrid algorithm that combines a reinforcement learning algorithm with an improved Arithmetic Optimization Algorithm (IAOA) which restructured mathematical optimization acceleration function. Specifically, Q-learning was employed to dynamically select different phases of Aquila Optimizer and IAOA for various problems, and experimental results showed that this hybrid algorithm exhibits high convergence speed and accuracy.

Moreover, some existing studies have utilized the BES algorithm for feature selection. The authors in [56] adopted chaotic local search and opposition-based learning to enhance search efficiency and mitigate the risk of the original BES algorithm being ensnared in local optima for feature selection. Additionally, in an enhanced BES [57], the authors utilized the entropy-based gradient to adjust the algorithm parameters to improve classification accuracy. Moreover, a hybrid chaos-based algorithm combining PSO and BES was proposed [58], where PSO was utilized to efficiently optimize the designed fitness function.

The algorithms described above can effectively tackle the challenge of feature selection across various domains. Nonetheless, However, subject to the NFL theorem, no single algorithm among these can universally tackle every feature selection challenge. Furthermore, same optimization algorithms may demonstrate varying efficacy across different feature selection scenarios. Therefore, we propose a novel IBES that balances exploration and utilization and maintains population diversity in an attempt to deal with a wider range of feature selection problems.

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3 Problem Formulation

In the feature selection process, each feature is either selected or discarded. Therefore, feature selection can be regarded as an optimization problem with a binary solution space. Assuming a dataset contains D features, feature $x_i (i = 1, \cdots, D)$ is represented by a binary number. Specifically, if the feature is selected, $x_i = 1$; otherwise, $x_i = 0$. Thus, the feature selection status of the dataset can be represented by a binary array $X = [x_1, x_2, \cdots, x_D]$, which constitutes a solution to the feature selection problem.

Given the dual considerations of the quantity of selected features and the accuracy of classification in feature selection, the problem inherently emerges as a multi-objective optimization challenge. Specifically, the first optimization objective is to reduce the dimensionality of the feature subset, which can be expressed as:

$$f_1 = \frac{D_s}{D}, \tag{1}$$

where D_s and D represent the selected and total features, respectively. Moreover, the second optimization objective is to reduce the error rate of classification, which can be denoted as:

$$f_2 = 1 - f_a, \tag{2}$$

where $f_a \in [0, 1]$ is the classification accuracy of classifier. Note that we use the k-nearest neighbor (KNN) algorithm as a classifier in this paper [59].

Considering the requirements of automated deployment and computational complexity of the algorithm, we apply the linear weighting method for handling the multi-objective optimization problem. Accordingly, the feature selection problem can be formulated as follows:

$$\begin{aligned} &\min_X F = (\delta \times f_1 + \lambda \times f_2), \\ &\text{s.t.}\, C_1 : x_i \in \{0, 1\}, \quad i = 1, 2, \cdots, D \\ &\qquad C_2 : 1 \le D_s \le D \\ &\qquad C_3 : 0 \le f_a \le 1 \end{aligned} \tag{3}$$

where F represents the fitness function. Moreover, $\delta \in [0, 1]$ and $\lambda = 1 - \delta$ are the weights of two objectives, respectively. In addition, C_1 represents the value range for each dimension of the solution. Moreover, C_2 and C_3 denote the constraints on the number of feature selections and the classification accuracy, respectively.

4 Proposed Method

4.1 Preliminaries of Bald Eagle Search

BES is a novel meta-heuristic search algorithm introduced by [25]. Since bald eagles have sharp vision and excellent tracking skills, BES aims to emulate the foraging strategies employed by bald eagles within natural ecosystems to solve optimization problems. Specifically, BES considers the individuals in the solution space as potential prey, while the process of searching solution space simulates hunting instincts of bald eagles. Furthermore, three stages are involved in the aforementioned process, including the selection stage, search stage, and swooping stage. In the following, BES will be shown in detail.

4.1.1 Selection Stage

During selection stage, bald eagles pinpoint and choose optimal regions within the designated search space. The process can be denoted as

$$P_{new,n} = P_b + \alpha \times rand \times (P_m - P_n), \tag{4}$$

where P represent the population and P_n is the n-th bald eagle. Moreover, $P_{new,n}$ denotes the updated position of n-th bald eagle. P_b denotes best position during the previous search. In addition, P_m denotes the average distribution of bald eagles. Moreover, α is utilized to regulate the movement of bald eagles, while $rand$ is a random number selected from $(0, 1)$.

4.1.2 Search Stage

During the search stage, while exploring the designated search space, bald eagles navigate in a spiral trajectory to expedite the process of locating prey. The process can be mathematically represented as

$$P_{new,n} = P_n + y(n) \times (P_n - P_{n+1}) + x(n) \times (P_n - P_m), \tag{5}$$

where P_{n+1} is the location of next bald eagle. Moreover, $x(n)$ and $y(n)$ correspond to the positions of n-th bald eagle within the polar coordinate system and take values in the range $[-1, 1]$ to change the spiral's shape, which can be denoted as

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$$x(n) = \frac{ds(n)}{max(|ds|)},$$
$$y(n) = \frac{dc(n)}{max(|dc|)},$$
$$ds(n) = d(n) \times \sin(\theta(n)),$$
$$dc(n) = d(n) \times \cos(\theta(n)), \quad (6)$$

where $\theta(n)$ and $d(n)$ indicate the polar angle and diameter of the spiral equation, which can be represented as

$$\theta(n) = b \times \pi \times r,$$
$$d(n) = \theta(n) + R \times r, \quad (7)$$

where b and R are set for controlling the spiral trajectory in the ranges of [5, 10] and [0.5, 2], respectively. It is worth noting that when b and R change, the algorithm augments its diversity, enabling it to evade local optima and consistently procure effective solutions. Moreover, r denotes the random number selected from (0, 1).

4.1.3 Swooping Stage

During the swooping stage, the bald eagle swings from the best position within the search space to the target prey. Moreover, all bald eagles converge upon the best position. The process can be denoted as

$$P_{new,n} = rand \times P_b + x_1(n) \times (P_n - k_1 \times P_m) + y_1(n)(P_n - k_2 \times P_b), \quad (8)$$

where k_1 and k_2 denote the intensity of movement exhibited by bald eagles directed to the optimal and central positions, respectively. Moreover, $x_1(n)$ and $y_1(n)$ are the polar coordinate equations to delineate the motion of these bald eagles during the swooping, which are represented as

$$x_1(n) = \frac{ds(n)}{max(|ds(n)|)},$$
$$y_1(n) = \frac{dc(n)}{max(|dc(n)|)},$$
$$ds(n) = d(n)) \times \sinh(\theta(n)),$$
$$dc(n) = d(n) \times \cosh(\theta(n)),$$
$$\theta(n) = b \times \pi \times r,\ d(n) = \theta(n). \quad (9)$$

4.2 Improved Bald Eagle Search

Although BES is easy to deploy and implement, it still face some of these limitations:

Algorithm 1: IBES

```
Initialize the population size N, and solution dimension D.
Initialize the numbers of iterations Iter_max.
Initialize the fitness function F.
Initialize the population P.
for Iter = 1 to Iter_max do
    for n = 1 to N do
        Update the new position P_{new,n} of n-th bald eagle in
        search stage according to Eq. (10).
        Calculate the fitness F(P_{new,n}) according to Eq. (3).
        if F(P_{new,n}) < F(P_b) then
            P_b = P_{new,n}
    for n = 1 to N do
        Update the new position P_{new,n} of n-th bald eagle in
        selection stage according to Eq. (14).
        Calculate the fitness F(P_{new,n}) according to Eq. (3).
        if F(P_{new,n}) < F(P_b) then
            P_b = P_{new,n}
    for n = 1 to N do
        Update the new position P_{new,n} of n-th bald eagle in
        swooping stage according to Eq. (8).
        Calculate the fitness F(P_{new,n}) according to Eq. (3).
        if F(P_{new,n}) < F(P_b) then
            P_b = P_{new,n}
    for n = 1 to N do
        Update the mutation position P_{new,n} of n-th bald eagle
        according to Eq. (17).
        Calculate the fitness F(P_{new,n}) according to Eq. (3).
        if F(P_{new,n}) < F(P_b) then
            P_b = P_{new,n}
```

- **Local Optimal**: The deficiency in global exploration capacity within original BES may lead to the premature convergence toward local optimal solutions.
- **Lack of diversity**: BES exhibits weakness in preserving the diversity of the population, hindering effective exploration of the solution space.

Therefore, we propose an IBES algorithm to alleviate the limitations above. The pseudo-code of IBES is shown in Algorithm 1.

4.2.1 Lévy Flight Mechanism

In the selection stage of BES, the search step size greatly influences the capability of exploring the solution space. Therefore, in order to broaden the search scope and provide BES with opportunities to escape local optima, we introduce the Lévy flight to enhance the selection stage of BES [60].

The Lévy flight entails a random search strategy following to the Lévy and heavy-tailed distribution, which realizes short-range search interspersed with occasional long-range walks. Leveraging the remarkable search capabilities inherent in the Lévy flight contributes to efficiently exploring solution space and evading entrapment in local optima. Therefore, the improved position update equation in (4) in

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the selection stage is rewritten as

$$P_{new,n} = P_b + \alpha \times r(P_m - P_n) \times Levy(\lambda), \tag{10}$$

where $Levy(\lambda) \sim \mu = t^{-\lambda}(1 < \lambda < 3)$ is a lévy flight factor. Due to the complexity of Lévy flight, it is usually modeled using the Mantegna algorithm, where the step size S is given by

$$S = \frac{\mu}{|v|^{\frac{1}{\beta}}}, \tag{11}$$

where $\mu \sim \mathcal{N}(0, \sigma^2)$ and $v \sim \mathcal{N}(0, 1)$. Moreover, σ can be denoted as

$$\sigma = \left\{ \frac{\tau(1+\beta)\sin(\frac{\pi\beta}{2})}{\tau(\frac{1+\beta}{2})\beta 2^{(\frac{\beta-1}{2})}} \right\}^{\frac{1}{\beta}}, \tag{12}$$

where τ is Gamma function, and β is set to 1.5.

4.2.2 Adaptive Weight

In the search stage, we introduce a novel adaptive weight operator for position adjustment process in (5). Specifically, during the early stage, a larger weight value can increase the search ranges of the bald eagles. During the later stage, a smaller weight value allows the bald eagles to conduct more detailed local explorations. Therefore, adopting the adaptive weight can balance the ability for global and local search, which contributes to enhancing the optimization capability and improving the convergence speed of BES. The adaptive weight can be represented as

$$\omega^t = \omega_{max} + (\omega_{max} - \omega_{min}) \times \sin(\pi + \frac{t}{Iter_{max}} \times (\frac{\pi}{2})), \tag{13}$$

where ω_{max} and ω_{min} denote the initial weight and end weight, respectively. Moreover, t represents the current number of iterations, while we denote $Iter_{max}$ as the maximum number of iterations. Accordingly, we rewrite the position equation in (5) as

$$P_{new,n} = P_n + \omega^t \times y(n) \times (P_n - P_{n+1}) + \omega^t \times x(n) \times (P_n - P_m). \tag{14}$$

4.2.3 Mutation Mechanism

As shown in Fig. 1, we propose a novel mutation mechanism to augment population diversity and enhance exploration capability for better solutions. Specifically, the mutation mechanism contains two strategies, which can be expressed as

- *Gaussian Mutation*: Gaussian mutation adopts random numbers obeying a normal distribution to modify the original position vector. In this case, most of the mutation operators are distributed around the original positions, effectively executing a neighborhood search. This mutation method not only enhances candidate solution diversity but also aids in evading local optimal regions. At the same time, a few operators are far away from the current position, which enhances the diversity of the population and is conducive to a better search of potential regions. Therefore, we adopt Gaussian mutation to allow bald eagles to search near the P_b to increase the chance of searching for superior solution. The Gaussian mutation process can be mathematically described as

$$P^G_{new,n} = P_b \times (1 + r_1), \tag{15}$$

where r_1 is Gaussian distribution.

- *Differential Mutation*: We deploy an improved differential mutation method to vary the position of the bald eagles, thereby enhancing the global search capability. The improved differential mutation method can be expressed as

$$P^D_{new,n} = p_1 r_2(P_b - P_n) + p_2 r_3(P_r - P_n), \tag{16}$$

where P_r is a random bald eagle selected from population. Moreover, p_1 and p_2 are the weight parameters. In addition, r_2 and r_3 are random numbers obeying Gaussian distribution.

Gaussian differential mutation perturbs individuals according to Eq. (16). Specifically, in the early stages of algorithm iteration, due to the uneven distribution of the population, there is a significant difference in the distribution of individual positions. Therefore, the algorithm primarily perturbs individuals through differential variables to generate diverse individuals. As the algorithm iterates, most positions in the population do not change significantly. At this stage, the algorithm mainly uses Gaussian distribution coefficients to perturb the population, thereby reducing the likelihood of getting trapped in local optima and avoiding premature convergence. Therefore, compared to the traditional differential mutation, the Gaussian differential mutation can generate larger perturbations for the current mutant individual, which makes it easier for the algorithm to escape local optima.

Consequently, the mutation mechanism can be denoted as

$$P_{new,n} = \begin{cases} P_n, & 0 < rand < R_1 \\ P^G_{new,n}, & R_1 \le rand < R_2 \\ P^D_{new,n}, & R_2 \le rand < 1 \end{cases} \tag{17}$$

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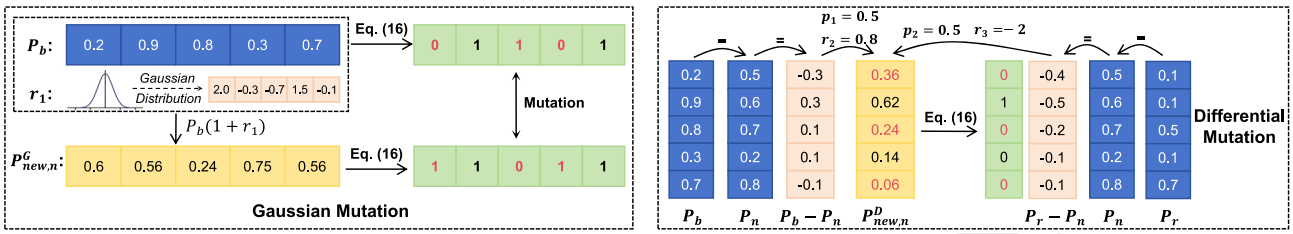


Fig. 1 Sketch map of mutation mechanism

where $rand \in (0, 1)$ is a random number. Moreover, R_1 and R_2 are the thresholds for Gaussian mutation and differential mutation, respectively. The proposed novel mutation mechanism posseses the ability to search around the global best and the current value, thus enhancing both population diversity and the exploratory capability of the algorithm.

4.2.4 IBES for Feature Selection

The proposed IBES is combined with the KNN classifier to form a wrapper mode for selecting features. Specifically, IBES is employed to train the dataset, seeking the most appropriate subset of features for KNN classifier. However, BES typically acts on continuous solution spaces, while the feature selection problem involves discrete variables (*i.e.*, whether to select a feature or not). Therefore, a traditional binary mechanism is implemented to map the solution from continuous space to discrete space [61], making the algorithm applicable to feature selection problems, which is delineated as follows:

$$P^b_{n,j} = \begin{cases} 1, & 0 < P_{n,j} < \eta \\ 0, & \eta \le P_{n,j} < 1 \end{cases} \tag{18}$$

where $P^b_{n,j} = 1$ indicates that the corresponding j-th feature is selected, otherwise, $P^b_{n,j} = 0$. In addition, we set the threshold η to 0.5. Consequently, the step-by-step process of IBES for feature selection is outlined as follows:

- **Step 1. Initialization**: In this step, the population is initialized according to the population size.
- **Step 2. Update Positions**: Note that the position update of the bald eagles occurs in three similar stages. Taking the selection stage as an example, the new positions of bald eagles are calculated according to Eq. (10), and then the positions are discretized by Eq. (18) to facilitate feature selection. Then, the corresponding fitness of new positions is calculated based on Eq. (3). Under the condition that the new position demonstrates superior performance compared to the original one, accept it and update the global best position; otherwise, keep the original position. Note that the evaluation and update processes for the search and swooping stages are the similar as the process described above, except that the position updates are performed according to Eqs. (10) and (8), respectively.
- **Step 3. Mutation Process**: The positions of bald eagles are mutated according to Eq. (17) for seeking the chance of escaping local optima, and then the positions are discretized by Eq. (18). Then, the corresponding fitness of new positions is calculated based on Eq. (3). Note that if the fitness of the new position surpasses the original and the global best position, their positions will be updated.
- **Step 4. Termination**: Repeat Step 1 to Step 3 until the maximum number of iterations is reached.
- **Step 5. Solution**: The best solution of IBES is returned, which represents the selected feature subset.
- **Step 6. Test**: The selected features are utilized to assess the performance of IBES across the datasets.

4.3 Complexity Analysis

We assume that the population size is N, the dimension of the search space is d, and the maximum number of iterations is T. Obviously, the complexity of population initialization stage is $\mathcal{O}(Nd)$. Additionally, the complexity of position updates in three stages is $\mathcal{O}(Nd)$, respectively. Therefore, the complexity of the original BES algorithm is $\mathcal{O}(Nd)+\mathcal{O}(T \cdot 3Nd)$. The proposed IBES algorithm performs an additional mutation stage compared to the original BES algorithm, and the complexity of this stage is $\mathcal{O}(Nd)$. Hence, the complexity of IBES is $\mathcal{O}(Nd) + \mathcal{O}(T \cdot 4Nd)$.

In conclusion, for the same number of iterations, the proposed IBES algorithm exhibits marginally greater time complexity compared to the BES algorithm. However, the optimization capability of our proposed algorithm outperforms BES, as corroborated through experimental validation presented in Sect. 5.

5 Experimental Results and Analysis

In this section, we construct tests to demonstrate the effectiveness of our proposed IBES algorithm.

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Table 1 Benchmark test function

No.	Test Function	n(Dim)	Range	Global Minimum
1	$f_1(x) = \sum_{i=1}^{n} x_i^2$	30	$[-100, 100]^n$	0
2	$f_2(x) = \sum_{i=1}^{n} \lvert x_i \rvert + \prod_{i=1}^{n} \lvert x_i \rvert$	30	$[-10, 10]^n$	0
3	$f_3(x) = \sum_{i=1}^{n} \left(\sum_{j=1}^{i} x_j \right)^2$	30	$[-100, 100]^n$	0
4	$f_4(x) = \max_i \{ \lvert x_i \rvert, 1 \le i \le d \}$	30	$[-100, 100]^n$	0
5	$f_5(x) = \sum_{i=1}^{n-1} \left[100(x_{i+1} - x_i^2)^2 + (x_i - 1)^2 \right]$	30	$[-30, 30]^n$	0
6	$f_6(x) = \sum_{i=1}^{n} (\lvert x_i + 0.5 \rvert)^2$	30	$[-100, 100]^n$	0
7	$f_7(x) = \sum_{i=1}^{n} i x_i^4 + random[0, 1)$	30	$[-1.28, 1.28]^n$	0
8	$f_8(x) = 418.9828n - \sum_{i=1}^{n} x_i \sin(\sqrt{\lvert x_i \rvert})$	30	$[-500, 500]^n$	0
9	$f_9(x) = \sum_{i=1}^{n} (x_i^2 - 10\cos(2\pi x_i) + 10)$	30	$[-5.12, 5.12]^n$	0
10	$f_{10}(x) = -20exp\left(-0.2\sqrt{(1/n)\sum_{i=1}^{n} x_i^2} \right) - exp\left((1/n)\sum_{i=1}^{n} \cos 2\pi x_i \right) + 20 + e$	30	$[-32, 32]^n$	0
11	$f_{11}(x) = 1 + \frac{1}{4000}\sum_{i=1}^{n} x_i^2 - \prod_{i=1}^{n} \cos\left(\frac{x_i}{\sqrt{i}} \right)$	30	$[-600, 600]^n$	0
12	$f_{12}(x) = \frac{\pi}{n}\{10\sin^2(\pi y_i) + \sum_{i=1}^{n-1}(y_i - 1)^2[1 + 10\sin^2(\pi y_{i+1})] + (y_n - 1)^2\} + \sum_{i=1}^{n} u(x_i, 10, 100, 4)$, $y_i = 1 + (x_i + 1)/4$, $u(x_i, a, k, m) = \begin{cases} k(x_i - a)^m & \text{if } x_i > a, \\ 0 & \text{if } -a \le x_i \le a, \\ k(-x_i - a)^m & \text{if } x_i < -a. \end{cases}$	30	$[-50, 50]^n$	0
13	$f_{13}(x) = 0.1\left[\sin^2(3\pi x_1) + \sum_{i=1}^{n-1}(x_i - 1)^2\left(1 + \sin^2(3\pi x_{i+1})\right) + (x_n - 1)^2\left(1 + \sin^2(2\pi x_n)\right)\right] + \sum_{i=1}^{n} u(x_i, 5, 100, 4)$	30	$[-50, 50]^n$	0
14	$f_{14}(x) = \left[\frac{1}{500} + \sum_{j=1}^{25} \frac{1}{j + \sum_{i=1}^{2}(x_i - a_{ij})^6} \right]^{-1}$	2	$[-65.536, 65.536]^n$	1
15	$f_{15}(x) = \sum_{i=1}^{11} \left[a_i - \frac{x_1(b_i^2 + b_i x_2)}{b_i^2 + b_i x_3 + x_4} \right]^2$	4	$[-5, 5]^n$	0.0003075
16	$f_{16}(x) = 4x_1^2 - 2.1x_1^4 + \frac{1}{3}x_1^6 + x_1x_2 - 4x_2^2 + 4x_2^4$	2	$[-5, 5]^n$	−1.0316285
17	$f_{17}(x) = (x_2 - \frac{5.1}{4\pi^2}x_1^2 + \frac{5}{\pi}x_1 - 6)^2 + 10(1 - \frac{1}{8\pi})\cos x_1 + 10$	2	$[-5, 10] \times [0, 15]$	0.398
18	$f_{18}(x) = \left[1 + (x_1 + x_2 + 1)^2 \left(19 - 14x_1 + 3x_1^2 - 14x_2 + 6x_1x_2 + 3x_2^2\right)\right] \times \left[30 + (2x_1 - 3x_2)^2 (18 - 32x_1 + 12x_1^2 + 48x_2 - 36x_1x_2 + 27x_2^2)\right]$	2	$[-2, 2]^n$	3
19	$f_{19}(x) = -\sum_{i=1}^{5} \left[(x - a_i)(x - a_i)^T + c_i \right]^{-1}$	4	$[0, 10]^n$	−10
20	$f_{20}(x) = -\sum_{i=1}^{4} c_i exp\left[-\sum_{j=1}^{4} a_{ij}(x_j - p_{ij}) \right]$	4	$[0, 1]^n$	−3.86
21	$f_{21}(x) = -\sum_{i=1}^{4} c_i exp\left[-\sum_{j=1}^{6} a_{ij}(x_j - p_{ij}) \right]$	6	$[0, 1]^n$	−3.32

5.1 Benchmark Test Functions

In order to verify the effectiveness of the improvements of our proposed IBES algorithm over the original BES algorithm, we introduce 21 benchmark test functions [62] as shown in Table 1. Each benchmark test function has different characteristics. Among them, f_1 to f_7 are unimodal functions, f_8 to f_{13} are multimodal functions, and f_{14} to f_{21} are fixed-dimension test functions.

5.2 Dataset of Feature Selection

We select 18 benchmark datasets from the UCI database to evaluate the performance of the proposed IBES algorithm and

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Table 2 Experimental results of BES and IBES on different benchmark test functions

F	Opt.	BES		IBES	
		Mean	Std	Mean	Std
f_1	0.00E+00	**0.00E+00**	**0.00E+00**	**0.00E+00**	**0.00E+00**
f_2	0.00E+00	**0.00E+00**	**0.00E+00**	**0.00E+00**	**0.00E+00**
f_3	0.00E+00	**0.00E+00**	**0.00E+00**	**0.00E+00**	**0.00E+00**
f_4	0.00E+00	**0.00E+00**	**0.00E+00**	**0.00E+00**	**0.00E+00**
f_5	0.00E+00	2.90E+01	**0.00E+00**	**2.81E+01**	7.88E-01
f_6	0.00E+00	6.43E+00	7.59E-01	**8.89E-09**	**2.53E-08**
f_7	0.00E+00	**7.32E-05**	**5.29E-05**	1.62E-04	1.25E-04
f_8	0.00E+00	8.41E+03	**5.61E+02**	**7.33E+03**	8.85E+02
f_9	0.00E+00	**0.00E+00**	**0.00E+00**	**0.00E+00**	**0.00E+00**
f_{10}	0.00E+00	**4.44E-16**	**0.00E+00**	**4.44E-16**	**0.00E+00**
f_{11}	0.00E+00	**0.00E+00**	**0.00E+00**	**0.00E+00**	**0.00E+00**
f_{12}	0.00E+00	9.20E-01	2.45E-01	**3.51E-10**	**1.18E-09**
f_{13}	0.00E+00	2.76E+00	**1.47E-01**	**1.07E+00**	1.21E+00
f_{14}	1.00E+00	**7.63E+00**	4.69E+00	12.67E+00	**3.00E+00**
f_{15}	3.075E-04	3.69E-03	8.98E-03	**3.47E-04**	**1.31E-04**
f_{16}	−1.0316285	**−1.03E+00**	4.44E-16	**−1.03E+00**	**3.36E-16**
f_{17}	3.98E-01	**3.98E-01**	**0.00E+00**	**3.98E-01**	**0.00E+00**
f_{18}	3.00E+00	3.89E+00	4.84E+00	**3.00E+00**	**3.11E-15**
f_{19}	−1.00E+01	**−5.33E+00**	1.44E+00	−5.05E+00	**9.31E-16**
f_{20}	−3.86E+00	**−3.86E+00**	1.20E-04	**−3.86E+00**	**6.06E-16**
f_{21}	−3.32E+00	−3.16E+00	1.58E-01	**−3.29E+00**	**5.25E-02**

other benchmark algorithms, most of which are commonly employed in feature selection tasks. Note that the details of the datasets are shown in Table 3.

5.3 Parameter Setting

We compare our proposed IBES algorithm with the standard BES algorithm and other state-of-the-art optimization algorithms such as Cuckoo Search (CS) [63], Firefly Algorithm (FA) [21], Genetic Algorithm (GA) [19], Particle Swarm Optimization (PSO) [23], Sparrow Search Algorithm (SSA) [64], ant colony optimization (ACO) [65], hybrid rice optimization (HRO) [66]. The parameter settings for each algorithm are shown in Table 4.

The population magnitude for all algorithms is established at 25, with the dimensionality of solutions aligning with the quantity of features within each dataset, and a restriction of 200 iterations is imposed. Additionally, in each trial, 80% of the instances are chosen for training, while the remaining 20% are used for testing, following a common approach utilized in various prior studies. Furthermore, each algorithm is executed independently for 30 iterations to mitigate the impact of randomness. Note that the primary metrics for assessing the optimization performance of the algorithms encompass three components: the fitness value, the count of selected features, and the classification accuracy.

5.4 Simulation Results on Benchmark Test Functions

The mean values and standard deviation (SD) values of the improved IBES algorithm and the original BES algorithm on the 21 benchmark test functions are shown in Table 2. Moreover, the best values acquired by a certain algorithm are highlighted using bold formatting for easier observation.

We can observe that the IBES algorithm achieves the best mean values on 18 functions and the best SD values on 17 functions. The results show that the improved algorithm outperforms the original BES algorithm in solving both high- and low-dimensional problems. In addition, the proposed IBES algorithm achieves the best mean values on six unimodal functions, which indicates that the IBES algorithm possesses strong local search capability and faster convergence speed. The performance above demonstrates the effectiveness of the improvement factors of IBES.

5.5 Simulation Results on Feature Selection

The performance of different algorithms on feature selection tasks is presented in this section.

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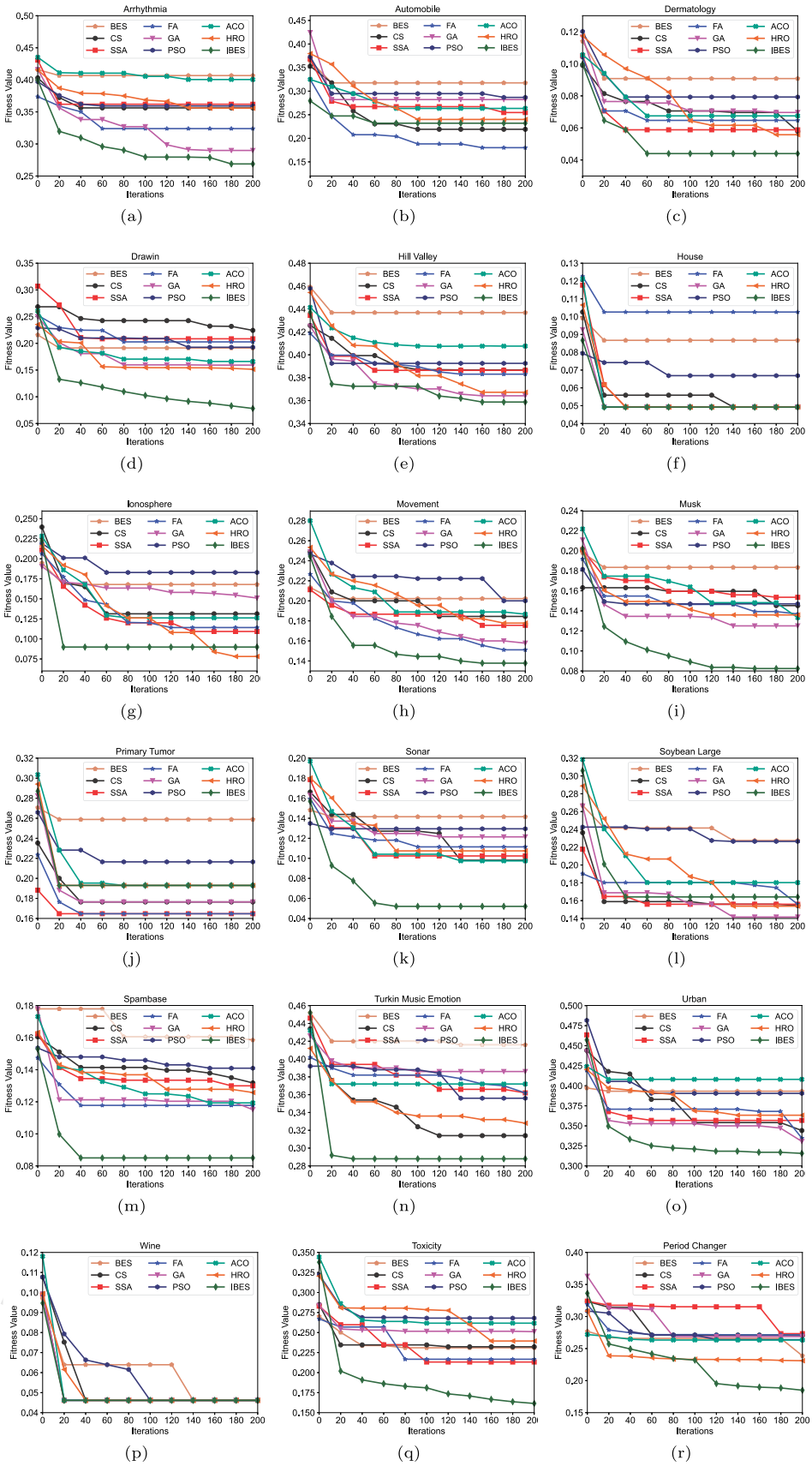


Fig. 2 Fitness convergence of different algorithms

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Table 3 Details of datasets

Number	Dataset	Number of Features	Number of Instances
1	Arrhythmia	279	452
2	Automobile	25	205
3	Dermatology	34	366
4	Drawnin	451	174
5	Hill-valley	100	606
6	House	16	435
7	Ionosphere	34	351
8	Movement	90	360
9	Musk	168	476
10	Primary Tumor	17	339
11	Sonar	60	208
12	Soybean Large	35	307
13	Spambase	57	460
14	Music Emotion	50	400
15	Urban	148	168
16	Wine	13	178
17	Toxicity	1203	171
18	Period Changer	1177	90

Table 4 Parameters of different algorithms

Algorithms	Parameters
BES	$\alpha = 1.5, b = 10, R = 1.5, k_1 = 1, k_2 = 1$ [25]
CS	discovery rate=0.25, step size=1 [67]
FA	attractiveness=1, absorption coefficient=1, step size=0.97 [68, 69]
GA	mutation rate=0.1, crossover rate=0.8 [70, 71]
PSO	$c_1 = 1.5, c_2 = 2.0$ [41]
SSA	alarm value=0.6, producer ratio=0.7, $SD = 0.2$ [64]
IBES	$\alpha = 1.5, b = 10, R = 1.5, k_1 = 1, k_2 = 1, \omega_{max} = 2, \omega_{min} = 0.5, R_1 = 0.5, R_2 = 0.75$ [25, 72, 73]
ACO	pheromone importance factor=1, heuristic pheromone importance factor=2, pheromone volatilization rate=0.5 [65]
ACO	crossbreeding probability=0.8, Maximum number of self-crosses=10 [66]

5.5.1 Evaluation of Fitness

The results showcasing the fitness values acquired by various algorithms are displayed in this section, which provides an intuitive demonstration of the performance of these algorithms. Tables 5 and 6 show the best fitness, worst fitness, mean fitness, SD values and CPU time of all algorithms over each dataset. Moreover, due to space constraints, the results of the chosen 18 datasets are delineated across two separate tables. We are able to observe that our proposed IBES algorithm outperforms the original BES algorithm on every dataset in terms of the best, the worst, the mean values. This indicates that the improvements proposed for BES are effective, which contributes to the ability of our proposed algorithm to develop globally and maintain population diversity.

Moreover, we can observe that the number of datasets for which the improved IBES algorithm outperforms the benchmark algorithms in terms of best, worst and mean fitness are 13, 13 and 16, respectively. In addition, we can observe that the selected 18 datasets contain some high-dimensional datasets (*e.g.*, Toxicity, Period Changer, and Drawin), and the IBES algorithm outperforms the other algorithms on these datasets, which suggests that IBES exhibits remarkable proficiency in addressing feature selection for high-dimensional datasets. Note that the excellent performance is attributed to the capability of global search, enabling IBES algorithm to comprehensively explore the feature space in high-dimensional datasets. Additionally, IBES consumes more CPU time when addressing feature selection problems on most datasets. This is due to the introduction of improvement factors that require additional computations on the solutions, resulting in increased experimental time. Although our proposed IBES algorithm demands slightly more runtime, the performance improvement brought by the improvement factors is crucial for many practical applications where data accuracy and reliability are prioritized over computational time.

In addition, the fitness value curves of the different algorithms on each dataset are shown in Fig. 2. We can observe

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Table 5 Fitness of different algorithms (Dataset 1 to 9)

Dataset	Algorithm	Best	Worst	Mean	SD	CPU Time
Arrhythmia	BES	0.3713	0.4166	0.3946	0.0135	531.2449
	CS	0.3298	0.3712	0.3508	**0.0098**	224.0094
	FA	0.2961	0.3635	0.3314	0.0175	1291.0605
	GA	0.2825	0.3563	0.3254	0.0218	**122.5356**
	PSO	0.3209	0.3882	0.3513	0.0157	124.5137
	SSA	0.3212	0.3628	0.3485	0.0097	128.8768
	ACO	0.3437	0.4188	0.3891	0.0155	128.3123
	HRO	0.3315	0.3755	0.3594	0.0128	248.2533
	IBES	**0.2553**	**0.3034**	**0.2801**	0.0146	467.0694
Automobile	BES	0.2633	0.4264	0.3337	0.0384	64.2269
	CS	0.1996	0.2901	0.2486	0.0212	35.9981
	FA	**0.1801**	0.3601	0.2597	0.0439	197.2051
	GA	0.1881	0.2901	0.2557	0.0202	17.7206
	PSO	0.2473	0.3870	0.2900	0.0354	**17.5915**
	SSA	0.1881	0.2706	0.2438	0.0209	17.8122
	ACO	0.2320	0.3495	0.2805	0.0242	24.6123
	HRO	0.2320	0.2633	0.2547	**0.0088**	49.9023
	IBES	0.2320	**0.2553**	**0.2382**	0.0091	66.1868
Dermatology	BES	0.0647	0.1232	0.0949	0.0141	95.5905
	CS	0.0471	0.0755	0.0612	0.0063	56.3654
	FA	**0.0411**	0.0941	0.0606	0.0105	298.5725
	GA	**0.0411**	0.0705	0.0558	0.0076	29.5409
	PSO	0.0529	0.0994	0.0811	0.0124	29.6686
	SSA	0.0519	0.0705	0.0598	0.0051	**29.5217**
	ACO	0.0498	0.0910	0.0649	0.0092	31.9356
	HRO	0.0498	0.0764	0.0616	0.0071	61.8052
	IBES	0.0439	**0.0529**	**0.0484**	**0.0044**	98.9402
Drawin	BES	0.1505	0.2693	0.2194	0.0269	702.7894
	CS	0.1583	0.2242	0.1902	0.0171	308.1528
	FA	0.1113	0.2027	0.1648	0.0231	1828.238
	GA	0.1201	0.1905	0.1501	0.0161	**150.9725**
	PSO	0.1225	0.2294	0.1688	0.0263	165.0162
	SSA	0.1579	0.2121	0.1891	**0.0146**	206.7354
	ACO	0.1208	0.2028	0.1627	0.0228	189.1825
	HRO	0.1070	0.1851	0.1474	0.0188	348.8123
	IBES	**0.0430**	**0.1313**	**0.0753**	0.0211	636.4161
Hill-valley	BES	0.4203	0.4458	0.4349	0.0062	160.1297
	CS	0.3781	0.4058	0.3905	0.0057	88.3373
	FA	0.3531	0.4061	0.3778	0.0121	494.0872
	GA	0.3565	0.3947	0.3755	0.0092	**43.7140**
	PSO	0.3672	0.4221	0.3983	0.0123	44.9253
	SSA	0.3716	**0.3938**	0.3833	**0.0052**	44.2378
	ACO	0.3990	0.4353	0.4191	0.0091	50.4543
	HRO	0.3672	0.4221	0.3935	0.0113	95.3837
	IBES	**0.3402**	0.4065	**0.3701**	0.0165	176.2749

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Table 5 continued

Dataset	Algorithm	Best	Worst	Mean	SD	CPU Time
House	BES	0.0616	0.1065	0.0811	0.0135	61.1965
	CS	**0.0491**	0.0617	0.0501	0.0031	34.7207
	FA	0.0492	0.1209	0.0642	0.0164	186.9045
	GA	**0.0491**	0.0617	0.0515	0.0047	**17.5295**
	PSO	**0.0491**	0.0918	0.0676	0.0113	17.5532
	SSA	**0.0491**	0.0492	0.0492	**0.0000**	27.4418
	ACO	**0.0491**	0.0616	0.0512	0.0046	24.9231
	HRO	**0.0491**	0.0616	0.0499	0.0028	47.5929
	IBES	**0.0491**	**0.0491**	**0.0491**	**0.0000**	86.4543
Ionosphere	BES	0.1620	0.2251	0.1964	0.0216	78.0641
	CS	0.0798	0.1543	0.1202	0.0184	52.0578
	FA	0.0627	0.1773	0.1138	0.0288	222.7640
	GA	**0.0568**	**0.1136**	0.0783	0.0159	24.0403
	PSO	0.0812	0.2010	0.1498	0.0351	**22.3926**
	SSA	0.0857	0.1425	0.1103	0.0147	25.7304
	ACO	0.0631	0.1946	0.1277	0.0341	35.2913
	HRO	0.0721	0.1502	0.1091	0.0201	64.2237
	IBES	0.0572	0.1229	**0.0761**	**0.0142**	86.5786
Movement	BES	0.1822	0.2308	0.2138	0.0154	130.8673
	CS	0.1711	0.1933	0.1826	**0.0052**	72.8738
	FA	0.1355	**0.1777**	0.1607	0.0101	432.0490
	GA	0.1244	0.1822	0.1602	0.0130	42.6032
	PSO	0.1666	0.2266	0.1950	0.0155	**38.6663**
	SSA	0.1577	0.1866	0.1774	0.0061	40.1216
	ACO	0.1555	0.2422	0.1958	0.0177	44.9876
	HRO	0.1600	0.2133	0.1874	0.0133	81.9227
	IBES	**0.1044**	0.1977	**0.1545**	0.0211	135.2023
Musk	BES	0.1608	0.2039	0.1833	0.0106	324.2172
	CS	0.1357	0.1642	0.1546	**0.0064**	160.3943
	FA	0.1119	0.1666	0.1392	0.0126	943.1506
	GA	0.1071	0.1607	0.1379	0.0129	87.5568
	PSO	0.1169	0.1782	0.1445	0.0153	**84.6080**
	SSA	0.1297	0.1714	0.1526	0.0094	87.1919
	ACO	0.1200	0.1782	0.1534	0.0153	96.6335
	HRO	0.1299	0.1719	0.1492	0.0107	182.1076
	IBES	**0.0764**	**0.1446**	**0.1020**	0.0164	275.2964

that IBES algorithm converges to the lowest fitness value on 14 datasets. Furthermore, IBES exhibits superior convergence speed compared to other benchmark algorithms across the majority of datasets. This indicates that IBES effectively balances exploitation and exploration, mitigating premature convergence issues by enhancing population diversity.

5.5.2 Classification Accuracy

The error rate of classification is shown in Tables 7 and 8. We can observe that the number of datasets for which the improved IBES algorithm outperforms the original algorithm in terms of mean, best, worst and SD values are 17, 16, 18 and 11, respectively. The error rate performance illustrates that the features identified through the IBES algorithm offer superior advantages for feature selection.

In addition, compared to other benchmark algorithms, IBES algorithm achieved the lowest mean error rate on 12 datasets, the lowest best error rate on 9 datasets, the lowest worst error rate on 9 datasets and the lowest SD value on 7 datasets. Moreover, SSA and FA outperform other algorithm over three dataset only. CS algorithm surpasses other

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Table 6 Fitness of different algorithms (Dataset 10 to 18)

Dataset	Algorithm	Best	Worst	Mean	SD	CPU Time
Primary Tumor	BES	0.2070	0.2752	0.2439	0.0178	53.1982
	CS	**0.1647**	0.1882	**0.1741**	0.0088	30.5588
	FA	**0.1647**	0.2352	0.1854	0.0162	166.7737
	GA	**0.1647**	0.2235	0.1827	0.0159	**15.5286**
	PSO	0.2047	0.2729	0.2329	0.0161	16.5131
	SSA	**0.1647**	**0.1764**	0.1694	0.0057	16.2752
	ACO	0.1929	0.2447	0.2042	0.0136	24.5858
	HRO	0.1929	0.2047	0.1950	0.0038	40.3443
	IBES	0.1929	0.1929	0.1929	**0.0000**	55.1771
Sonar	BES	0.1107	0.1748	0.1474	0.0160	73.5445
	CS	0.0957	0.1247	0.1112	**0.0075**	52.5196
	FA	0.0600	0.1181	0.0959	0.0154	268.2965
	GA	0.0690	0.1371	0.1025	0.0146	29.3917
	PSO	0.0753	0.1569	0.1158	0.0185	**27.1693**
	SSA	0.0923	0.1261	0.1102	0.0076	28.1982
	ACO	0.0653	0.1328	0.0998	0.0147	33.807
	HRO	0.0874	0.1394	0.1049	0.0110	57.7772
	IBES	**0.0366**	**0.0994**	**0.0725**	0.0132	106.3715
Soybean Large	BES	0.1942	0.3002	0.2516	0.0278	82.0173
	CS	0.1301	0.1646	0.1526	0.0077	51.0162
	FA	**0.1172**	0.1859	**0.1443**	0.0166	299.3549
	GA	0.1301	0.2033	0.1475	0.0134	**24.4606**
	PSO	0.1828	0.2690	0.2216	0.0200	29.48955
	SSA	0.1273	0.2529	0.1487	**0.0073**	27.9374
	ACO	0.1698	**0.1601**	0.2144	0.0213	33.4343
	HRO	0.1433	0.2161	0.1832	0.0167	56.3125
	IBES	0.1397	0.2264	0.1724	0.0188	86.9184
Spambase	BES	0.1228	0.1690	0.1522	0.0092	177.8018
	CS	0.1185	0.1344	0.1284	0.0040	129.9107
	FA	0.1002	0.1369	0.1187	0.0090	756.3862
	GA	0.1037	0.1344	0.1172	0.0079	76.3556
	PSO	0.1130	0.1494	0.1313	0.0092	**65.5816**
	SSA	0.1204	0.1353	0.1285	**0.0039**	67.6561
	ACO	0.1143	0.1515	0.1358	0.0092	105.0235
	HRO	0.1075	0.1382	0.1250	0.0071	213.5239
	IBES	**0.0849**	**0.1045**	**0.0926**	0.0059	445.2181
Turkish Music Emotion	BES	0.3480	0.4480	0.4106	0.0245	78.9158
	CS	0.2760	0.3820	0.3372	0.0259	50.9762
	FA	0.2739	0.3980	0.3510	0.0371	276.0700
	GA	0.3080	0.4060	0.3745	0.0189	30.6191
	PSO	0.3120	0.4320	0.3869	0.0297	**25.3285**
	SSA	0.3080	0.3860	0.3581	0.0192	26.2938
	ACO	0.3600	0.4360	0.4025	**0.0174**	29.5532
	HRO	0.2800	0.3920	0.3398	0.0283	62.0612
	IBES	**0.2400**	**0.3520**	**0.3157**	0.0309	117.7459

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Table 6 continued

Dataset	Algorithm	Best	Worst	Mean	SD	CPU Time
Urban	BES	0.3931	0.4693	0.4241	0.0202	203.6744
	CS	0.2698	0.3680	0.3368	0.0198	98.1942
	FA	0.2576	0.4038	0.3346	0.0316	609.2287
	GA	0.3015	0.4079	0.3516	0.0279	50.1579
	PSO	0.2938	0.4394	0.3894	0.0264	54.8483
	SSA	0.3056	0.3694	0.3378	0.0162	56.1906
	ACO	0.3537	0.4353	0.3909	0.0181	**30.0213**
	HRO	0.3442	0.3918	0.3649	**0.0122**	115.6725
	IBES	**0.2040**	**0.3510**	**0.3083**	0.0297	210.1206
Wine	BES	**0.0461**	0.0970	0.0745	0.0155	39.8716
	CS	**0.0461**	**0.0461**	**0.0461**	**0.0000**	23.7314
	FA	**0.0461**	0.0905	0.0559	0.0141	134.6311
	GA	**0.0461**	0.0615	0.0466	0.0027	**12.4133**
	PSO	**0.0461**	0.0639	0.0477	0.0048	24.2475
	SSA	**0.0461**	**0.0461**	**0.0461**	**0.0000**	16.2217
	ACO	**0.0461**	0.0615	0.0476	0.0046	23.9713
	HRO	**0.0461**	**0.0461**	**0.0461**	**0.0000**	36.3527
	IBES	**0.0461**	**0.0461**	**0.0461**	**0.0000**	40.3595
Toxicity	BES	0.2095	0.2670	0.2524	0.0135	2286.2985
	CS	0.2085	0.2562	0.2298	0.0147	1089.5442
	FA	0.2127	0.2565	0.2369	0.0121	5182.2081
	GA	0.1795	0.2632	0.2321	0.0239	**502.3644**
	PSO	0.1937	0.2917	0.2526	0.0270	510.1498
	SSA	0.2093	0.2529	0.2260	**0.0113**	1633.3057
	ACO	0.2216	0.2869	0.2655	0.0166	683.3128
	HRO	0.2048	0.2756	0.2435	0.0193	1178.47
	IBES	**0.1472**	**0.2196**	**0.1845**	0.0182	1798.0148
Period Changer	BES	0.2308	0.2692	0.2507	0.0155	1721.0797
	CS	0.2668	0.3139	0.2792	0.0151	797.1432
	FA	0.2607	0.3165	0.2771	0.0199	4478.9256
	GA	0.2623	0.3587	0.3168	0.0270	**412.3410**
	PSO	0.2254	0.3023	0.2636	0.0235	413.1385
	SSA	0.2703	0.3148	0.2836	0.0166	1205.6527
	ACO	0.2375	0.3028	0.2658	**0.0123**	551.0683
	HRO	0.2196	0.2673	0.2425	0.0158	1015.43
	IBES	**0.1247**	**0.2238**	**0.1911**	0.0195	1360.4655

algorithms on two dataset. GA, ACO, and HRO outperform other algorithms on just one datasets. The performance above demonstrates that our proposed IBES can obtain better classification accuracy while maintaining the stability of exploration. The enhancements introduced in the factors could potentially balance both exploration and exploitation capabilities, consequently improving the performance.

5.5.3 Number of Selected Features

The number of selected features of different algorithms over each dataset is shown in Tables 9 and 10. We can find that IBES algorithm outperforms the original BES algorithm in terms of mean, best and worst number of selected features over all dataset, respectively. Moreover, the number of datasets in which the IBES algorithm outperforms other benchmark algorithms in terms of mean, best, worst and SD values are 16, 14, 17 and 6, respectively. It is notable that only SSA algorithms achieve the best mean number of selected

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Table 7 Error rate of classification (Dataset 1 to 9)

Dataset	Algorithm	Best	Worst	Mean	SD
Arrhythmia	BES	0.3451	0.4159	0.3775	0.0180
	CS	0.2967	0.3516	0.3252	0.0128
	FA	**0.2527**	0.3406	0.3031	0.0207
	GA	0.2747	0.4065	0.3205	0.0253
	PSO	0.3097	0.3893	0.3404	0.0204
	SSA	0.2967	0.3406	0.3245	**0.0119**
	ACO	0.3185	0.4070	0.3707	0.0191
	HRO	0.3097	0.3628	0.3392	0.0163
	IBES	0.2673	**0.3362**	**0.3011**	0.0171
Automobile	BES	0.2692	0.4230	0.3384	0.0445
	CS	**0.1951**	0.2926	0.2544	0.0232
	FA	**0.1951**	0.3902	0.2666	0.0470
	GA	**0.1951**	0.2926	0.2682	0.0208
	PSO	0.2500	0.4038	0.3025	0.0403
	SSA	**0.1951**	0.2926	**0.2528**	0.0263
	ACO	0.2500	0.3461	0.2814	0.0207
	HRO	0.2500	0.3076	0.2653	0.0125
	IBES	0.2500	**0.2692**	0.2538	**0.0076**
Dermatology	BES	**0.0000**	0.0326	0.0170	0.0095
	CS	**0.0000**	0.0270	0.0067	0.0083
	FA	**0.0000**	0.0270	0.0060	0.0071
	GA	**0.0000**	0.0270	0.0058	0.0075
	PSO	**0.0000**	0.0434	0.0141	0.0112
	SSA	**0.0000**	0.0270	0.0076	0.0075
	ACO	**0.0000**	0.0326	0.0094	0.0078
	HRO	**0.0000**	0.0217	0.0094	0.0078
	IBES	**0.0000**	**0.0108**	**0.0054**	**0.0054**
Drawin	BES	0.0681	0.2272	0.1560	0.0350
	CS	0.0857	0.1714	0.1209	0.0204
	FA	0.0285	0.1428	0.0933	0.0312
	GA	0.0285	0.1142	0.0733	**0.0190**
	PSO	0.0454	0.1818	0.1113	0.0328
	SSA	0.0857	0.1428	0.1171	0.0200
	ACO	0.0454	0.1363	0.0871	0.0269
	HRO	**0.0227**	**0.1136**	0.0719	0.0235
	IBES	**0.0227**	0.1363	**0.0583**	0.0266
Hill-valley	BES	0.4078	0.4473	0.4267	**0.0103**
	CS	0.3770	0.4180	0.3920	0.0108
	FA	0.3524	**0.4016**	0.3781	0.0117
	GA	0.3606	0.4098	0.3827	0.0114
	PSO	0.3815	0.4276	0.4024	0.0115
	SSA	0.3688	0.4180	0.3967	0.0117
	ACO	0.4013	0.4407	0.4208	0.0112
	HRO	0.3815	0.4276	0.4006	0.0127
	IBES	**0.3421**	0.4078	**0.3774**	0.0141

Table 7 continued

Dataset	Algorithm	Best	Worst	Mean	SD
House	BES	0.0366	0.0550	0.0431	0.0053
	CS	0.0459	0.0459	0.0459	**0.0000**
	FA	**0.0229**	0.0574	**0.0417**	0.0091
	GA	**0.0229**	0.0459	0.0452	0.0041
	PSO	0.0275	**0.0458**	0.0434	0.0046
	SSA	0.0459	0.0459	0.0459	**0.0000**
	ACO	0.0458	**0.0458**	0.0458	**0.0000**
	HRO	0.0275	**0.0458**	0.0452	0.0032
	IBES	0.0458	**0.0458**	0.0458	**0.0000**
Ionosphere	BES	0.1363	0.2045	0.1651	0.0182
	CS	0.0704	0.1408	0.1051	0.0165
	FA	0.0563	0.1408	0.0962	0.0272
	GA	**0.0422**	0.1126	0.0685	0.0177
	PSO	0.0681	0.1818	0.1265	0.0271
	SSA	0.0704	0.1267	0.0948	0.0162
	ACO	0.0568	0.1477	0.1121	0.0282
	HRO	0.0568	0.1363	0.0984	0.0197
	IBES	0.0568	**0.1022**	**0.0674**	**0.0117**
Movement	BES	0.1444	0.1999	0.1707	0.0111
	CS	0.1111	0.1527	0.1301	**0.0098**
	FA	0.0972	**0.1388**	0.1138	0.0115
	GA	0.0777	**0.1388**	0.1196	0.0136
	PSO	0.1222	0.1777	0.1477	0.0143
	SSA	0.1111	0.1999	0.1314	0.0099
	ACO	0.1222	0.1527	0.1529	0.0182
	HRO	0.1222	0.1777	0.1459	0.0124
	IBES	**0.0694**	0.1423	**0.1115**	0.0181
Musk	BES	0.0924	0.1344	0.1142	0.0123
	CS	0.0521	0.1041	0.0836	**0.0105**
	FA	0.0312	0.0937	0.0656	0.0159
	GA	0.0416	0.0937	0.0681	0.0125
	PSO	0.0420	0.1092	0.0756	0.0151
	SSA	0.0521	0.1041	0.0829	0.0130
	ACO	0.0504	0.1092	0.0826	0.0184
	HRO	0.0588	0.1092	0.0778	0.0137
	IBES	**0.0336**	**0.0840**	**0.0537**	0.0149

features on four datasets, while the other benchmark algorithms either achieve the best mean on only one dataset or fail to achieve the best mean.

Moreover, we can observe that IBES excels particularly in feature selection for high-dimensional datasets (*e.g.*, Toxicity, Period Changer, and Drawin), where the number of selected features is typically much lower compared to other algorithms. Furthermore, as seen by combining the statistics data in Tables 7 and 8, IBES achieves significant feature reduction without compromising classification accuracy, and

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Table 8 Error rate of classification (Dataset 10 to 18)

Dataset	Algorithm	Best	Worst	Mean	SD
Primary Tumor	BES	0.1764	0.3058	0.2254	0.0270
	CS	0.1617	**0.1911**	0.1828	0.0129
	FA	**0.1471**	0.2205	**0.1799**	0.0172
	GA	**0.1471**	0.2058	0.1799	0.0164
	PSO	0.1764	0.2823	0.2274	0.0243
	SSA	0.1617	**0.1911**	0.1833	0.0130
	ACO	0.1764	0.2470	0.2101	0.0127
	HRO	0.1764	0.2235	0.2070	0.0123
	IBES	0.2117	0.2117	0.2117	**0.0000**
Sonar	BES	0.0384	0.1153	0.0756	0.0191
	CS	0.0238	0.0714	0.0468	0.0114
	FA	**0.0000**	**0.0476**	0.0373	0.0170
	GA	0.0238	0.0714	0.0436	0.0107
	PSO	0.0192	0.0961	0.0532	0.0176
	SSA	0.0238	0.0952	0.0523	0.0142
	ACO	0.0192	0.0769	0.0391	0.0126
	HRO	0.0192	0.0769	0.0416	**0.0111**
	IBES	**0.0000**	0.0576	**0.0307**	0.0127
Soybean Large	BES	0.1298	0.2597	0.1948	0.0374
	CS	0.0483	0.1129	0.0779	0.0195
	FA	**0.0322**	**0.0967**	**0.0704**	0.0201
	GA	0.0483	0.1612	0.0698	0.0233
	PSO	0.1038	0.2077	0.1584	0.0240
	SSA	0.0483	0.1129	0.0768	**0.0170**
	ACO	0.0779	0.1948	0.1437	0.0282
	HRO	0.0649	0.1818	0.1138	0.0289
	IBES	0.0389	0.1688	0.1091	0.0308
Spambase	BES	0.0834	0.1259	0.0974	0.0098
	CS	0.0705	0.1237	0.0877	0.0114
	FA	0.0694	0.1085	0.0852	0.0081
	GA	0.0694	0.1085	0.0859	0.0078
	PSO	0.0729	0.1285	0.0932	0.0105
	SSA	**0.0640**	0.1118	0.0855	0.0102
	ACO	0.0755	0.1146	0.0929	0.0091
	HRO	0.0695	0.1103	0.0877	0.0097
	IBES	0.0703	**0.0886**	**0.0793**	**0.0049**
Turkish Music Emotion	BES	0.3299	0.4700	0.4121	0.0348
	CS	0.2500	**0.4125**	**0.3304**	0.0396
	FA	0.2500	0.4250	0.3679	0.0595
	GA	0.2750	0.4375	0.4143	0.0298
	PSO	0.2900	0.4700	0.4106	0.0455
	SSA	0.2750	0.4125	0.3550	0.0403
	ACO	0.3399	0.4399	0.4146	**0.0271**
	HRO	0.2800	0.4399	0.3520	0.0519
	IBES	**0.2299**	0.4300	0.3419	0.0675

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Table 8 continued

Dataset	Algorithm	Best	Worst	Mean	SD
Urban	BES	0.3809	0.4761	0.4142	0.0257
	CS	**0.2352**	**0.3529**	0.3049	0.0245
	FA	**0.2352**	0.3823	0.3147	0.0416
	GA	0.2647	0.3823	0.3235	0.0387
	PSO	0.2857	0.4523	0.3952	0.0334
	SSA	0.2647	**0.3529**	**0.3009**	0.0236
	ACO	0.3333	0.4285	0.3730	0.0232
	HRO	0.3333	0.3809	0.3571	**0.0162**
	IBES	0.2380	0.4047	0.3547	0.0338
Wine	BES	**0.0000**	0.0444	0.0162	0.0151
	CS	**0.0000**	**0.0000**	**0.0000**	**0.0000**
	FA	**0.0000**	0.0555	0.0064	0.0137
	GA	**0.0000**	**0.0000**	**0.0000**	**0.0000**
	PSO	**0.0000**	0.0222	0.0007	0.0039
	SSA	**0.0000**	**0.0000**	**0.0000**	**0.0000**
	ACO	**0.0000**	**0.0000**	**0.0000**	**0.0000**
	HRO	**0.0000**	**0.0000**	**0.0000**	**0.0000**
	IBES	**0.0000**	**0.0000**	**0.0000**	**0.0000**
Toxicity	BES	0.1395	0.2093	0.1937	0.0173
	CS	0.1428	**0.1999**	0.1666	0.0196
	FA	0.1428	0.2000	0.1741	0.0174
	GA	**0.1142**	**0.1999**	0.1723	0.0290
	PSO	0.1395	0.2558	0.2116	0.0346
	SSA	0.1428	**0.1999**	0.1639	**0.0153**
	ACO	0.1627	0.2325	0.2116	0.0202
	HRO	0.1395	0.2325	0.1875	0.0254
	IBES	0.1162	0.2093	**0.1615**	0.0243
Period Changer	BES	0.1739	**0.2173**	0.1942	0.0216
	CS	0.2222	0.2777	0.2314	0.0207
	FA	0.2222	0.2777	0.2345	0.0230
	GA	0.2222	0.3333	0.2796	0.0335
	PSO	0.1739	0.2608	0.2159	0.0307
	SSA	0.2222	0.2777	0.2351	0.0234
	ACO	0.1739	0.2608	0.2130	**0.0172**
	HRO	0.1739	**0.2173**	0.1913	0.0212
	IBES	**0.0869**	**0.2173**	**0.1782**	0.0259

even attains the highest mean accuracy. The improved performance suggests that IBES is able to explore the most critical regions of the search space while avoiding irrelevant regions, which may be attributed to the Lévy flight mechanism and the mutation mechanism. The search space in high-dimensional datasets is extremely large, and traditional methods are prone to getting stuck in local optima. Lévy flight allows for more effective exploration of the entire space, ensuring that the proposed algorithm can escape local optima and find more representative feature subsets. Moreover, The introduction of mutation mechanism is particularly important in high-dimensional datasets as it increases the diversity of solutions and thus helps the proposed algorithm to reduce redundant features. Hence, thanks to the above improvement factors, IBES enhances classification accuracy while decreasing the count of chosen features through preserving the most informative features.

5.5.4 Wilcoxon Signed-Rank Test

The Wilcoxon signed-rank test is a non-parametric statistical hypothesis test [74], which is typically utilized to determine

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Table 9 Number of selected features (Dataset 1 to 9)

Dataset	Algorithm	Best	Worst	Mean	SD
Arrhythmia	BES	113.0000	147.0000	129.2000	8.8859
	CS	107.0000	150.0000	126.3666	8.2198
	FA	103.0000	146.0000	126.3666	8.2198
	GA	**27.0000**	130.0000	96.3333	32.8799
	PSO	96.0000	121.0000	110.2666	6.0988
	SSA	101.0000	146.0000	124.0666	8.6561
	ACO	117.0000	143.0000	129.0333	6.6054
	HRO	104.0000	140.0000	122.8000	8.5455
	IBES	39.0000	**58.0000**	**48.0000**	**4.7046**
Automobile	BES	4.0000	12.0000	7.8666	2.0121
	CS	**2.0000**	9.0000	5.6333	1.7603
	FA	3.0000	10.0000	5.8000	1.7204
	GA	3.0000	9.0000	5.1333	1.5434
	PSO	4.0000	9.0000	6.0000	1.2110
	SSA	3.0000	7.0000	5.2000	1.2489
	ACO	4.0000	11.0000	6.9333	1.9482
	HRO	**2.0000**	7.0000	5.3000	1.1000
	IBES	4.0000	**5.0000**	**4.4000**	**0.4898**
Dermatology	BES	10.0000	18.0000	13.8333	2.0669
	CS	7.0000	12.0000	9.5000	1.5000
	FA	7.0000	16.0000	9.7666	1.8741
	GA	7.0000	12.0000	8.7000	1.7269
	PSO	8.0000	16.0000	11.8666	1.9102
	SSA	**6.0000**	12.0000	9.1333	**1.2036**
	ACO	7.0000	14.0000	9.7666	1.7641
	HRO	**6.0000**	13.0000	9.2000	1.8511
	IBES	**6.0000**	**9.0000**	**7.5000**	1.5000
Drawin	BES	186.0000	232.0000	212.9333	10.0097
	CS	192.0000	246.0000	210.3666	10.7718
	FA	178.0000	227.0000	202.8666	11.3188
	GA	184.0000	223.0000	205.7000	10.0371
	PSO	154.0000	203.0000	179.4333	12.8444
	SSA	199.0000	240.0000	214.7000	**8.9485**
	ACO	186.0000	238.0000	209.4666	11.0385
	HRO	187.0000	226.0000	202.2666	9.1612
	IBES	**47.0000**	**84.0000**	**64.4333**	9.1603
Hill-valley	BES	36.0000	52.0000	43.7666	4.3103
	CS	33.0000	47.0000	38.4333	**3.7030**
	FA	28.0000	49.0000	37.6666	4.8808
	GA	**22.0000**	**42.0000**	34.6666	4.5411
	PSO	25.0000	46.0000	38.2333	4.7306
	SSA	25.0000	43.0000	**32.9666**	4.7852
	ACO	33.0000	54.0000	41.2333	4.4172
	HRO	29.0000	45.0000	36.5000	4.1372
	IBES	24.0000	48.0000	34.1333	5.3897

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Table 9 continued

Dataset	Algorithm	Best	Worst	Mean	SD
House	BES	2.0000	6.0000	3.7333	1.1234
	CS	**1.0000**	2.0000	1.0666	0.2494
	FA	**1.0000**	6.0000	2.4666	1.3349
	GA	**1.0000**	3.0000	1.2333	0.4955
	PSO	**1.0000**	5.0000	2.6333	1.0796
	SSA	**1.0000**	**1.0000**	**1.0000**	**0.0000**
	ACO	**1.0000**	**2.0000**	**1.1666**	0.3726
	HRO	**1.0000**	**3.0000**	**1.1000**	0.3958
	IBES	**1.0000**	**1.0000**	**1.0000**	**0.0000**
Ionosphere	BES	7.0000	16.0000	10.9333	2.1281
	CS	3.0000	10.0000	6.1333	1.7651
	FA	3.0000	11.0000	6.2666	1.9652
	GA	**2.0000**	**7.0000**	4.0000	1.1547
	PSO	3.0000	13.0000	8.2666	2.8559
	SSA	**2.0000**	9.0000	5.8666	1.3349
	ACO	**3.0000**	13.0000	6.4666	2.5394
	HRO	**2.0000**	9.0000	5.1666	1.5723
	IBES	**2.0000**	**7.0000**	**3.7666**	**1.1160**
Movement	BES	27.0000	45.0000	37.8000	4.4676
	CS	27.0000	42.0000	35.3666	3.5259
	FA	26.0000	38.0000	31.3333	3.4189
	GA	21.0000	39.0000	31.9666	4.3472
	PSO	29.0000	40.0000	34.5666	**2.8364**
	SSA	26.0000	39.0000	32.5333	3.1804
	ACO	26.000	41.0000	33.0666	3.5017
	HRO	23.0000	40.0000	31.8333	4.1156
	IBES	**16.0000**	**35.0000**	**26.4666**	4.3492
Musk	BES	66.0000	86.0000	77.2000	**4.9084**
	CS	58.0000	91.0000	73.6666	6.9586
	FA	57.0000	87.0000	72.9000	7.4982
	GA	58.0000	84.0000	70.1333	6.0151
	PSO	56.0000	91.0000	70.6333	7.4138
	SSA	63.0000	81.0000	72.4666	4.9580
	ACO	59.0000	83.0000	73.3666	6.5344
	HRO	59.0000	85.0000	73.0000	5.7965
	IBES	**26.0000**	**68.0000**	**49.5666**	10.2198

whether the distributions of two related samples are identical. Therefore, to further examine the performance differences between the IBES algorithm and other comparison algorithms, a Wilcoxon signed-rank test is performed, as shown in Table 11. We observe that the sum of positive ranks S^+ is significantly higher than the sum of negative ranks S^- in all comparisons. Furthermore, all p values are less than the significance level $\alpha = 0.05$, indicating a significant difference between the IBES algorithm and the other comparison algorithms. Therefore, the IBES algorithm is effective in feature selection.

5.5.5 Performance Comparison Between Multi-objective Algorithms and Single-Objective IBES

We introduce several multi-objective optimization algorithms, *i.e.*, multi-objective PSO (MOPSO), multi-objective evolutionary algorithm based on decomposition (MOEA/D), multi-objective simulated annealing (MOSA), and NSGA-II, to compare with the multi-objective versions of the original bald eagle search (BES) algorithm and our proposed IBES algorithm, referred to as MOBES and MOIBES, respectively.

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Table 10 Number of selected features (Dataset 10 to 18)

Dataset	Algorithm	Best	Worst	Mean	SD
Primary Tumor	BES	2.0000	8.0000	5.4000	1.5832
	CS	**1.0000**	5.0000	2.3666	1.4715
	FA	**1.0000**	7.0000	3.5333	1.7074
	GA	**1.0000**	8.0000	3.3000	1.8823
	PSO	**1.0000**	8.0000	4.3333	1.4452
	SSA	**1.0000**	4.0000	**1.9333**	1.2892
	ACO	2.0000	6.0000	3.0666	1.1234
	HRO	2.0000	5.0000	2.5000	1.0246
	IBES	2.0000	**2.0000**	2.0000	**0.0000**
Sonar	BES	19.0000	34.0000	26.0666	3.6872
	CS	18.0000	29.0000	22.1333	2.3342
	FA	17.0000	24.0000	19.8333	**1.9336**
	GA	15.0000	25.0000	20.3000	2.6476
	PSO	18.0000	30.0000	22.0000	2.8982
	SSA	14.0000	25.0000	20.5000	2.9636
	ACO	14.0000	27.0000	20.5666	3.1057
	HRO	17.0000	28.0000	21.5000	2.3629
	IBES	**11.0000**	**18.0000**	**14.3666**	1.9576
Soybean Large	BES	12.0000	24.0000	16.7666	2.8128
	CS	12.0000	22.0000	15.8000	2.2715
	FA	11.0000	21.0000	15.4000	2.4711
	GA	11.0000	**19.0000**	16.0333	**1.8345**
	PSO	12.0000	21.0000	16.6000	1.8726
	SSA	11.0000	**19.0000**	15.2666	2.3084
	ACO	12.0000	21.0000	17.4000	2.3748
	HRO	**10.0000**	20.0000	16.1333	2.4458
	IBES	**10.0000**	**19.0000**	**14.9000**	2.5475
Spambase	BES	16.0000	25.0000	21.1666	2.3816
	CS	9.0000	21.0000	16.6000	2.7760
	FA	10.0000	17.0000	14,4000	**1.5187**
	GA	10.0000	20.0000	13.8000	2.4276
	PSO	11.0000	23.0000	16.1666	2.6593
	SSA	10.0000	23.0000	17.1333	2.5785
	ACO	12.0000	23.0000	17.5333	2.7535
	HRO	10.0000	21.0000	15.6333	2.5753
	IBES	**6.0000**	**11.0000**	**8.3000**	1.7916
Turkish Music Emotion	BES	13.0000	30.0000	19.8000	3.4775
	CS	12.0000	26.0000	18.2333	3.6577
	FA	8.0000	23.0000	14.1666	4.2667
	GA	5.0000	22.0000	11.2333	3.5840
	PSO	5.0000	22.0000	14.6000	**2.3921**
	SSA	10.0000	27.0000	18.5333	4.3721
	ACO	12.0000	26.0000	17.7000	3.5038
	HRO	5.0000	24.0000	14.5666	4.8626
	IBES	**2.0000**	**20.0000**	**10.5333**	6.6319

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Table 10 continued

Dataset	Algorithm	Best	Worst	Mean	SD
Urban	BES	59.0000	76.0000	68.1666	4.8654
	CS	56.0000	82.0000	68.3333	6.7593
	FA	40.0000	78.0000	60.9000	9.5196
	GA	50.0000	85.0000	68.2000	6.9205
	PSO	44.0000	71.0000	53.8333	5.9165
	SSA	64.0000	80.0000	71.3333	**4.2921**
	ACO	56.0000	80.0000	68.0333	5.6479
	HRO	46.0000	71.0000	58.2666	6.0494
	IBES	**9.0000**	**26.0000**	**18.0666**	5.2594
Wine	BES	**3.0000**	6.0000	4.0000	0.9309
	CS	**3.0000**	**3.0000**	**3.0000**	**0.0000**
	FA	**3.0000**	5.0000	3.3000	0.5859
	GA	**3.0000**	4.0000	3.0333	0.1795
	PSO	**3.0000**	4.0000	3.0666	0.2494
	SSA	**3.0000**	**3.0000**	**3.0000**	**0.0000**
	ACO	**3.0000**	4.0000	3.1000	0.3000
	HRO	**3.0000**	**3.0000**	**3.0000**	**0.0000**
	IBES	**3.0000**	**3.0000**	**3.0000**	**0.0000**
Toxicity	BES	556.0000	615.0000	585.9666	14.8559
	CS	549.0000	610.0000	580.8000	14.3559
	FA	554.0000	616.0000	587.4761	15.4384
	GA	530.0000	628.0000	566.6666	24.8989
	PSO	446.0000	537.0000	501.6333	21.6617
	SSA	557.0000	612.0000	580.5000	**13.0326**
	ACO	545.0000	622.0000	578.7000	19.5075
	HRO	527.0000	602.0000	562.1666	17.5595
	IBES	**287.0000**	**377.0000**	**322.8333**	22.4886
Period Changer	BES	524.0000	607.0000	561.3000	18.7761
	CS	524.0000	592.0000	553.5333	**17.9382**
	FA	488.0000	555.0000	526.7777	20.7245
	GA	498.0000	770.0000	548.3666	47.8849
	PSO	496.0000	583.0000	534.7333	21.3009
	SSA	531.0000	610.0000	562.1333	20.4364
	ACO	520.0000	598.0000	561.2333	18.3151
	HRO	474.0000	577.0000	526.5666	23.6779
	IBES	**247.0000**	**335.0000**	**285.5333**	21.6575

We select four datasets (*i.e.*, Automobile, Sonar, Musk, Period Changer) with feature dimensions ranging from low to high to validate the performance of these algorithms, and the Pareto sets of the aforementioned algorithms on these datasets are shown in Fig. 3. It can be seen from the figures that the Pareto set obtained by MOIBES is more closer to the direction of Pareto front in both smaller-scale and larger-scale datasets, which indicates that MOIBES outperforms other multi-objective algorithms in solving the feature selection problem across different dimensions of the datasets. This is due to the lévy flight mechanism that enhances the global search capability of MOIBES and the mutation mechanism that enhances the diversity of the population, which allows the MOIBES algorithm to better explore the solution space. Moreover, combining the results obtained by IBES using linear weighting as shown in Tables 7-10, we can observe that the solutions obtained by IBES are not significantly dominated by the Pareto solutions from any multi-objective algorithm, which demonstrates the effectiveness of the linear weighting approach in solving the feature selection problem.

In addition, we conduct further analysis from the perspective of fitness values to compare the multi-objective

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Table 11 Results of Wilcoxon signed-rank test

Algorithms		S^+	S^-	p value	+	≈	−	Decision
IBES	BES	171	0	**0.000**	18	0	0	+
	CS	144	9	**0.001**	15	1	2	+
	FA	158	13	**0.002**	16	0	2	+
	GA	154	17	**0.003**	16	0	2	+
	PSO	171	0	**0.000**	18	0	0	+
	SSA	140	13	**0.003**	15	1	2	+
	ACO	171	0	**0.000**	18	0	0	+
	HRO	153	0	**0.0000**	17	1	0	+

algorithms with the proposed single-objective IBES. Specifically, for each multi-objective algorithm, we selected two representative solutions from its Pareto set: one with the minimum number of selected features and one with the lowest classification error rate. We calculated their fitness values f_C and f_E using the linear weighting approach shown in Eq. 3. These fitness values f_C and f_E are compared with the fitness values of the single-objective IBES, and the results are shown in Table 12. We find that the fitness values obtained by the IBES algorithm are superior to those of the multi-objective optimization algorithms. This is because the IBES algorithm directly focuses on minimizing the combined metric after linearly weighting the two objectives, with the search process explicitly guided toward solutions that provide the best trade-offs according to the specified weights.

5.5.6 Performance Analysis of Binary Mechanism

We compared the sigmoid binary mechanism with our adopted traditional binary threshold mechanism, and the results are shown in Table 13. Although the Sigmoid mechanism is widely used in feature selection problems, the traditional binary threshold mechanism performs better in the proposed algorithm. This may be due to the fact that the IBES algorithm already incorporates lévy flight and mutation mechanisms, which provide sufficient randomness and exploration capability in the solution space. Introducing additional randomness during the binary mapping phase may result in excessive perturbations, thereby interfering with convergence of the proposed algorithm. Moreover, compared to the Sigmoid mechanism, the binary threshold mechanism better preserves meaningful improvements in the solution space, simplifies decision boundaries, and enhances computational efficiency.

5.5.7 Performance Analysis of Population Size

The performance of the proposed IBES algorithm for different population sizes are presented in Table 14. We observe that as the population size increases, the fitness value, error rate, and the number of selected features decrease accordingly. This is because the larger population allows the algorithm to maintain more candidate solutions in each generation, expanding the coverage of the search space and increasing solution diversity. Therefore, the larger population helps reduce the risk of getting trapped in local optima, thereby improving the feature selection performance of the algorithm. However, the runtime increases with population size. This is because the larger population requires evaluating and updating more individuals in each generation, which increases computational overhead and extends the runtime.

5.5.8 Performance Analysis of Improvement Factors

In this part, we conduct experiments to assess the efficacy of the enhancement factors in the IBES algorithm. Specifically, we address the feature selection task by utilizing original BES algorithm, BES with adaptive weighting factor (BES weight) and BES with mutation mechanism (BES mutation), BES with lévy flight mechanism (BES-lévy) to evaluate the performance of these factors. Note that the algorithms above are tested on four selected datasets, *i.e.*, Arrhythmia, Sonar, Dermatology and Wine, where the dimensionality of these datasets ranges from high to low. We present the experimental results acquired by the aforementioned algorithms in Table 15.

Performance of BES Weight. We can observe that BES weight outperforms original BES algorithm in the low and middle-dimensional dataset, *i.e.*, Dermatology and Sonar. This stems from the proposed adaptive weighting factor, which can adjust the search range of the algorithm, thereby enhancing the exploratory capacity in the initial phase of the BES algorithm. However, the adaptive weighting factor leads the BES algorithm to focus on local search during the later stages, potentially constraining its performance. Evidently, preserving population diversity is vital to address this limitation. Hence, introducing additional improvement factors becomes imperative.

Performance of BES Mutation. The BES mutation algorithm demonstrates superior performance over the original algorithm, as evidenced by enhancements in the fitness evaluation, accuracy in classification, and the quantity of selected features. This is due to the fact that the mutation mechanism maintains the diversity of the population. Specifically, the Gaussian mutation facilitates the exploration of the vicinity of the global optimum, which aids in escaping local optima. In addition, the difference mutation gives the algorithm more powerful local exploration capabilities. Therefore, the proposed mutation mechanism steers the algorithm toward exploring around high-quality solutions, which enhances the mining power of BES algorithm.

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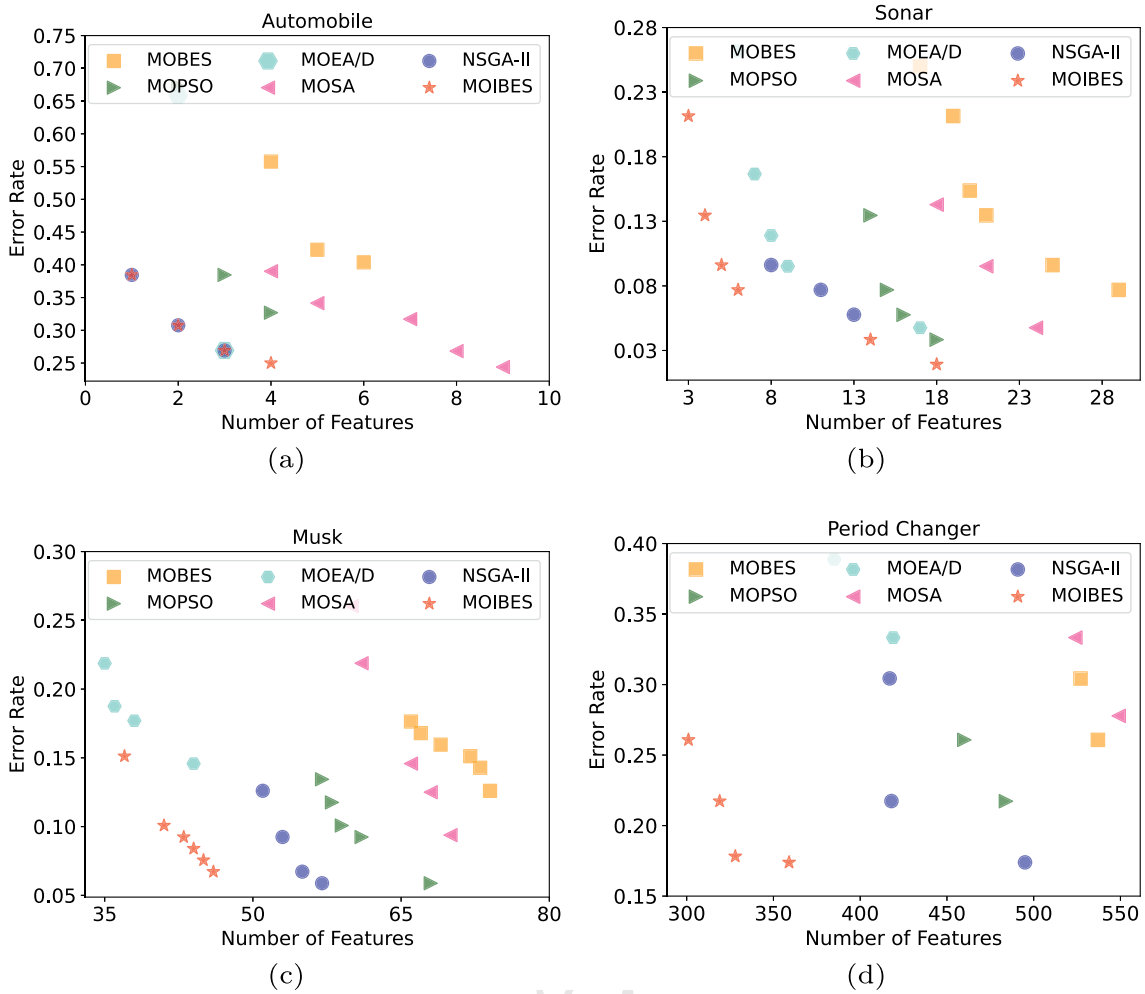


Fig. 3 The Pareto set of different multi-objectives algorithms

Table 12 Fitness comparison of multi-objective algorithms and IBES

Algorithms	Datasets							
	Automobile		Sonar		Musk		Period Changer	
MOBES	f_C	0.4780	f_C	0.2556	f_C	0.2196	f_C	0.3329
	f_E	0.3710	f_E	0.1581	f_E	0.1888	f_E	0.2998
MOPSO	f_C	0.5544	f_C	0.1543	f_C	0.1753	f_C	0.2868
	f_E	0.2935	f_E	0.0907	f_E	0.1279	f_E	0.2560
MOEA/D	f_C	0.5428	f_C	0.2295	f_C	0.2166	f_C	0.3764
	f_E	0.2393	f_E	0.0947	f_E	0.1690	f_E	0.3378
MOSA	f_C	0.3441	f_C	0.1743	f_C	0.2797	f_C	0.3556
	f_E	0.2671	f_E	0.1180	f_E	0.1583	f_E	0.3155
NSGA-II	f_C	0.3156	f_C	0.1035	f_C	0.1615	f_C	0.3142
	f_E	0.2393	f_E	0.0894	f_E	0.1148	f_E	0.2232
MOIBES	f_C	0.3156	f_C	0.1792	f_C	0.1650	f_C	0.2597
	f_E	0.2320	f_E	0.0753	f_E	0.1085	f_E	0.2001
IBES	0.2382		0.0725		0.1020		0.1911	

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Table 13 The performance comparison between Sigmoid mapping method and our adopted method

Binary mechanism	Metrics	Datasets			
		Automobile	Sonar	Musk	Period Changer
Sigmoid	Fitness	0.2400	0.0904	0.1143	0.1992
	Error rate	0.2557	0.0358	0.0717	0.1768
	Count	4.4333	18.5333	47.8666	339.9666
Our	Fitness	0.2382	0.0725	0.1020	0.1911
	Error rate	0.2538	0.0307	0.0537	0.1782
	Count	4.4000	14.3666	49.5666	285.5333

Table 14 The performance analysis of different population size

Datasets	Population Size	Mean Fitness	Mean Error Rate	Mean Count	CPU Time
Automobile	10	0.2483	0.2653	4.5333	36.3724
	25	0.2382	0.2538	4.4000	66.1868
	50	0.2381	0.2557	4.2000	104.0975
	75	0.2350	0.2532	4.0666	157.9032
Sonar	10	0.0777	0.0397	13.8333	58.2852
	25	0.0725	0.0307	14.3666	106.3715
	50	0.0708	0.0307	13.8666	173.3604
	75	0.0701	0.0307	13.6666	267.8996
Musk	10	0.1117	0.0546	57.1333	89.2063
	25	0.1020	0.0537	49.5666	275.2964
	50	0.0928	0.0407	46.4666	575.4214
	75	0.0871	0.0389	47.0666	638.4209
Period Changer	10	0.2071	0.1884	332.2333	578.9249
	25	0.1911	0.1782	285.5333	1360.4655
	50	0.1820	0.1739	252.4000	2869.7942
	75	0.1767	0.1710	235.2666	4246.3012

Table 15 The performance analysis of different improvement factors

Datasets	Metrics	BES	BES weight	BES mutation	BES lévy
Arrhythmia	Fitness	0.3946	0.4009	0.3521	0.2959
	Error rate	0.3775	0.3840	0.3324	0.3156
	Count	130.7000	130.0000	120.300	60.3000
Sonar	Fitness	0.1474	0.1446	0.0962	0.0797
	Error rate	0.0765	0.0733	0.0384	0.3840
	Count	26.0666	25.8000	19.6333	14.5000
Dermatology	Fitness	0.0949	0.0897	0.0525	0.0530
	Error rate	0.0170	0.0161	0.0083	0.0086
	Count	13.8333	13.0666	7.8000	7.8300
Wine	Fitness	0.0745	0.0733	0.0461	0.0461
	Error rate	0.0162	0.0140	0.0000	0.0000
	Count	4.0000	4.0333	3.0000	3.0000

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Performance of BES-lévy. The BES-lévy algorithm demonstrates superior performance in terms of fitness value, classification accuracy, and the number of selected features compared to the original algorithm, which are attributed to the integration of the lévy flight mechanism. Lévy flight mechanism is a stochastic walk method, and the stochastic exploration helps the BES algorithm to effectively get rid of local optima, which are a common pitfall in optimization tasks. Moreover, lévy flight mechanism enhances the global search capability of BES algorithm significantly by introducing long-distance jumps, increasing the exploration range, thereby improving algorithm performance on high-dimensional datasets.

In summary, these three improvement factors are effective in enhancing the performance of the original BES. Furthermore, combining some previous statistical data, we can observe that using any one of them alone cannot achieve the performance of IBES. Therefore, they are complementary. For instance, solely employing adaptive weighting factors on high-dimensional datasets might cause the algorithm to become trapped in local optima. Therefore, it becomes imperative to incorporate both the Lévy flight mechanism and mutation mechanism to mitigate this issue.

6 Conclusion

In this paper, we propose a novel IBES algorithm aimed at solving the feature selection problem. Specifically, we first introduce the Lévy flight mechanism to improve the global search efficiency of the algorithm. Second, an adaptive weighting operator is introduced to balance the global and local search capability. In addition, we employ the mutation mechanism to augment the population diversity of the algorithm. The proposed algorithm is experimentally evaluated on several typical datasets, and the numerical results present that IBES outperforms BES, CS, FA, GA, PSO, and SSA algorithms in solving the feature selection problem. Moreover, we find that our proposed IBES algorithm excels in handling the feature selection problem on high-dimensional datasets.

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Author Contributions JF: validation, software, writing-original draft. XZ: conceptualization, methodology. LZ: writing-review&editing.

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Data Availability All dataset used in experiment are available on public platform.

Declarations

Conflict of Interest No conflict of interest exits in the submission of this manuscript.

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